

Module 09 Sequential Data Access

IBM InfoSphere Information Server for Data Integration



Module Objectives

After completing this topic, you should be able to:

- Describe different types of file data
- Describe the Reject Link
- Read data using Multiple Readers in Sequential files
- Explain Data Set usage and properties
- Handle NULL values in data
- Explain HDFS/BDFS architecture, usage and properties

Types of File Data Access

- Sequential File
 - Can be variable or fixed length
- Data Set
 - Contains a control file (.ds) and one or more data files (not readable)
- BDFS / File connector
 - Big Data File System. Exchange information with Big Data Sources

Sequential Data Process Flow

- Import and export operators are generated
 - Stages get translated into operators during the compile
- Import operators convert data from the external format, as described by the Table Definition, to the framework internal format
 - Internally, the format of data is described by schemas
- Export operators reverse the process
- Messages in the job log use the “import” / “export” terminology
 - e.g., “100 records imported successfully; 2 rejected”
 - e.g., “100 records exported successfully; 0 rejected”
 - Records get rejected when they cannot be converted correctly during the import or export

Sequential File Stage

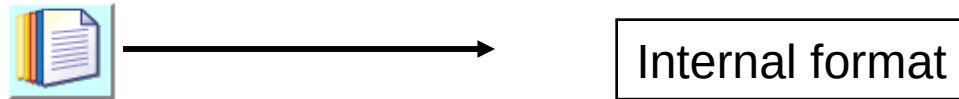


- Used to read data from or write data to one or more flat (sequential) files
- Can have a single input link or a single output link, and a single reject link
- Supports various formats (.txt, .csv)
- Normally executes in sequential mode
- Executes in parallel when reading multiple files
- Can use multiple readers
 - E.g. 4 Readers create 4 partitions from a single file; each partition read sequentially
- The stage needs to be told:
 - How file is divided into rows (record format)
 - How row is divided into columns (column format)

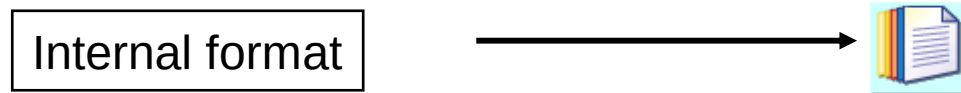
Using the Sequential File Stage

- Both import and export of sequential files (text, binary) can be performed.

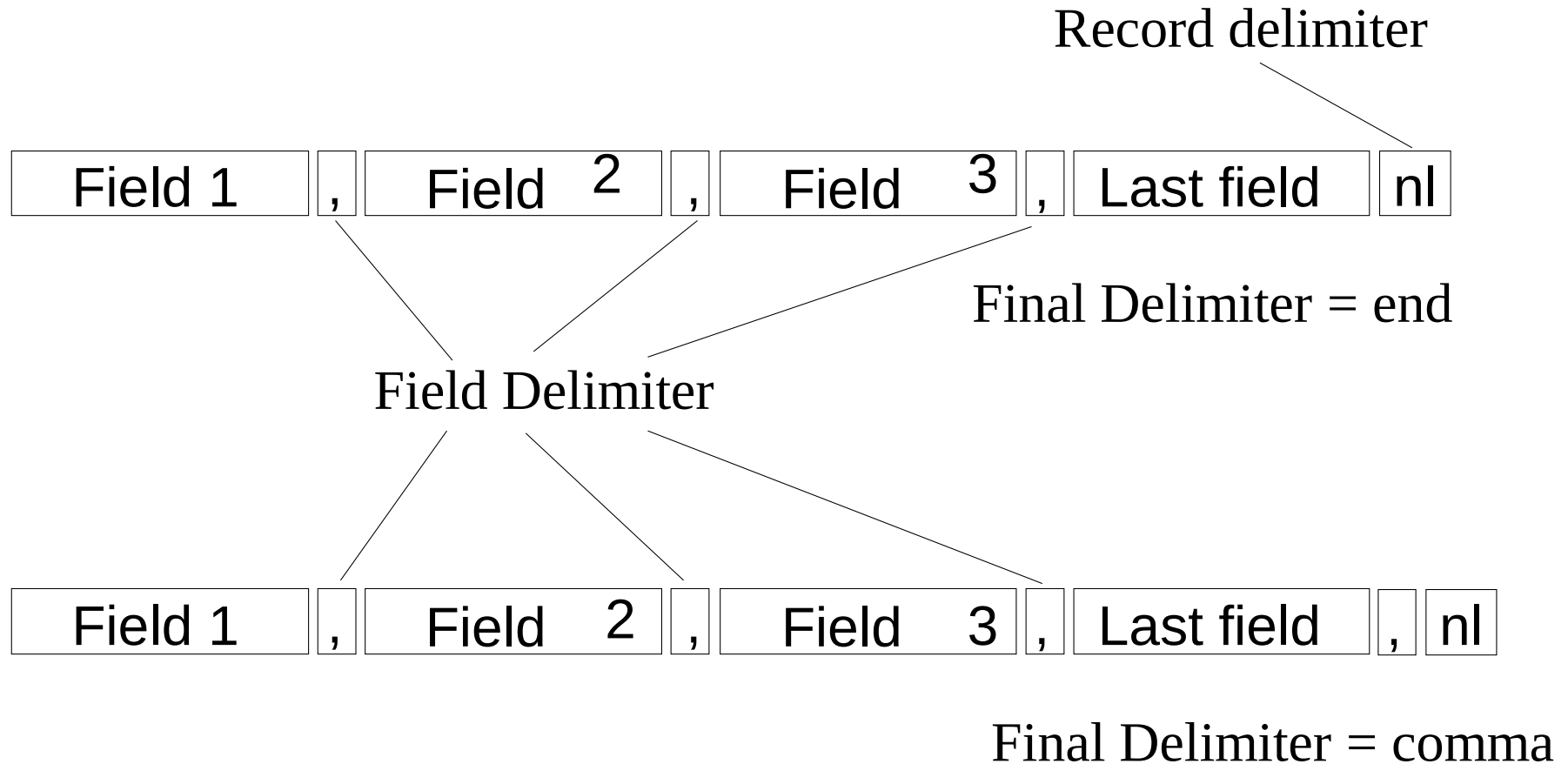
- Data import:



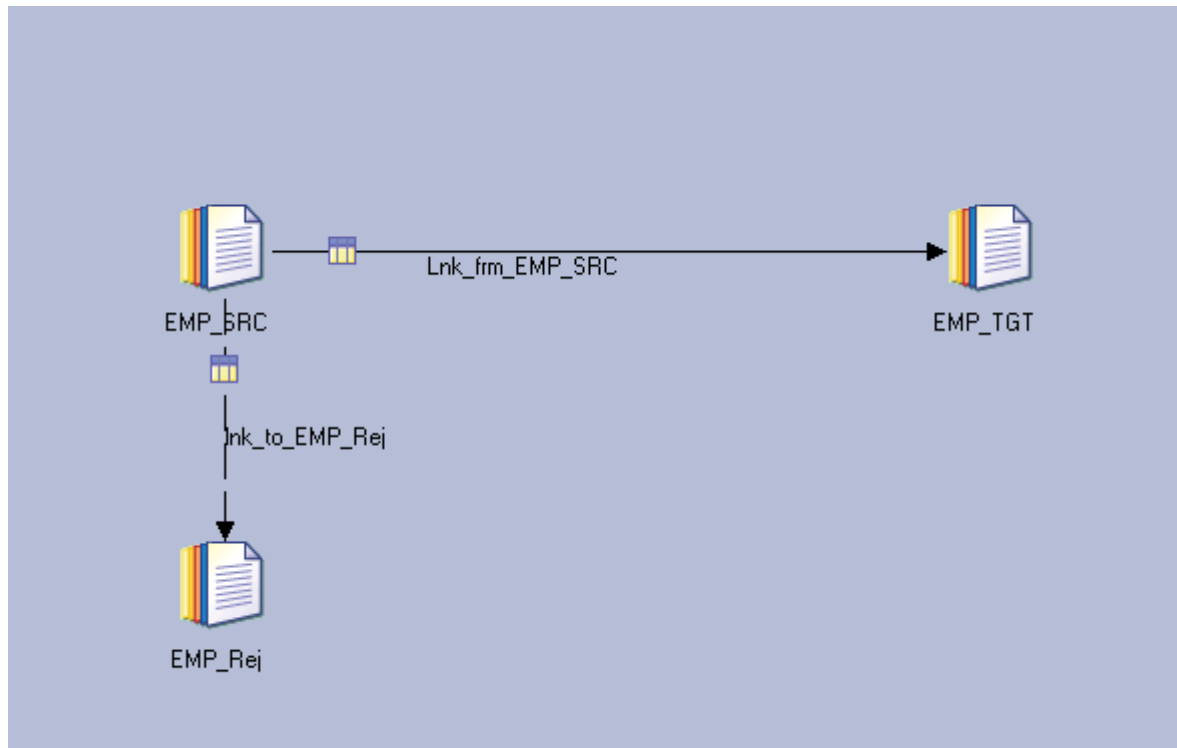
- Data export:



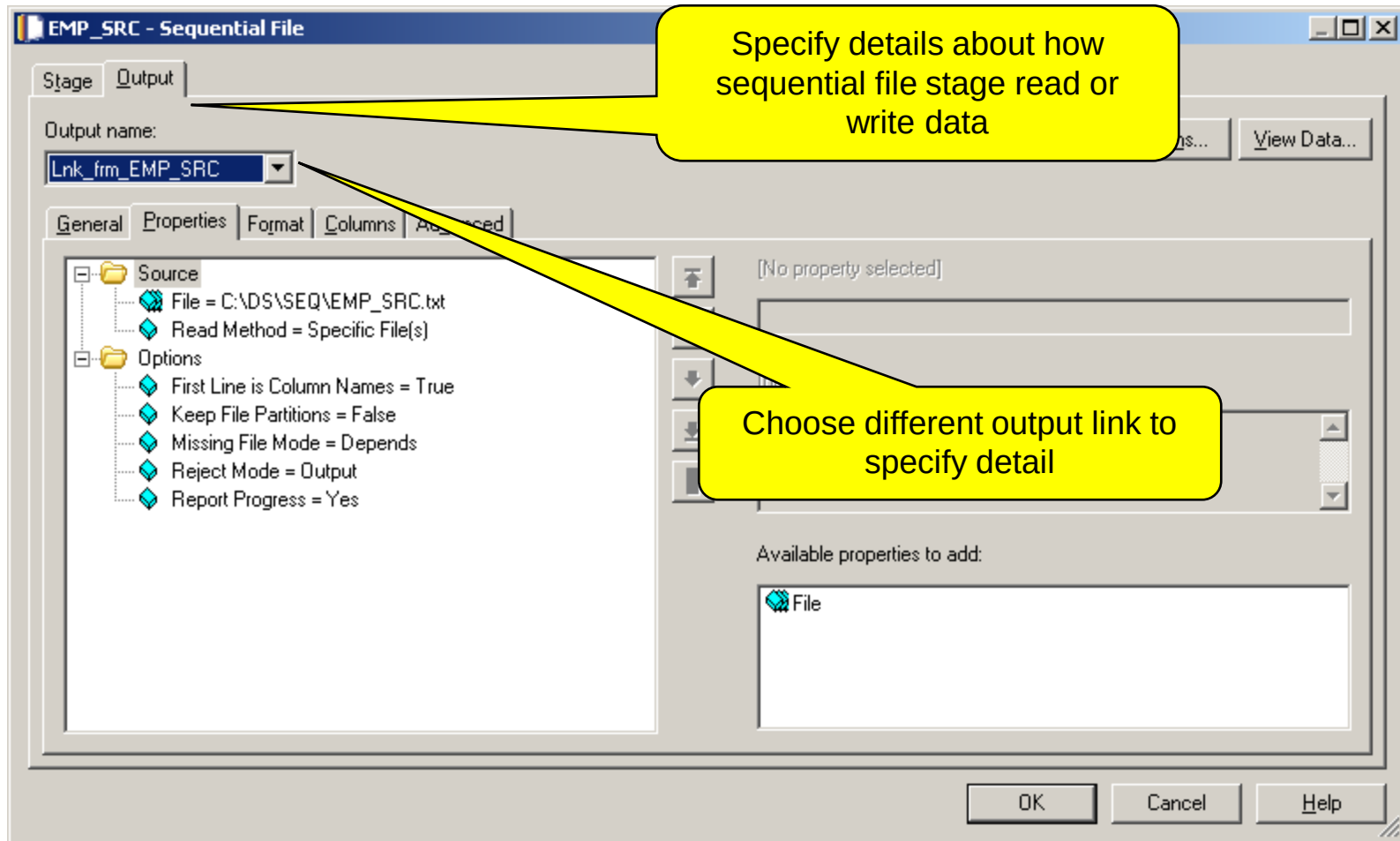
Sequential File Format Example



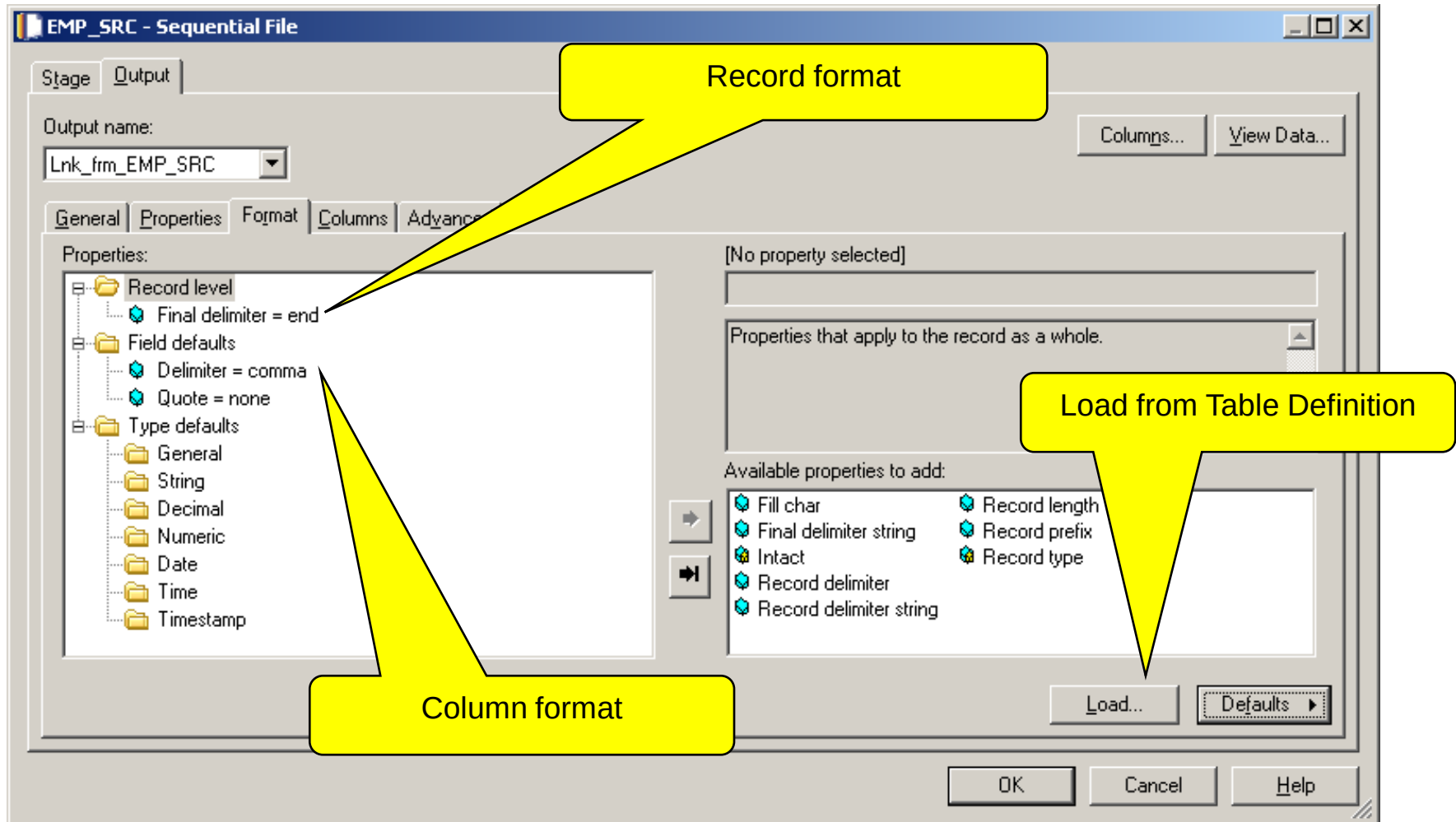
Sequential File Stage - Job Example



Sequential File Stage - Properties



Sequential File Stage - Format Tab



Sequential File Stage - Columns Tab

EMP_SRC - Sequential File

Stage Output

Output name:
Lnk_frm_EMP_SRC

Columns... View Data...

General Properties Format Columns Advanced

	Column name	Key	SQL type	Length	Scale	Nullable	Description
1	EMPNO	☐	Integer	10		No	
2	ENAME	☐	VarChar	255		No	
3	JOB	☐	VarChar	255		No	
4	MGR	☐	Integer	10		No	
5	HIREDATE	☐	Date	10		No	
6	SAL	☐	Numeric	10		No	
7	COMM	☐	Numeric	10		No	
8	DEPTNO	☐	Integer	10		No	

View data

Load from Table Definition

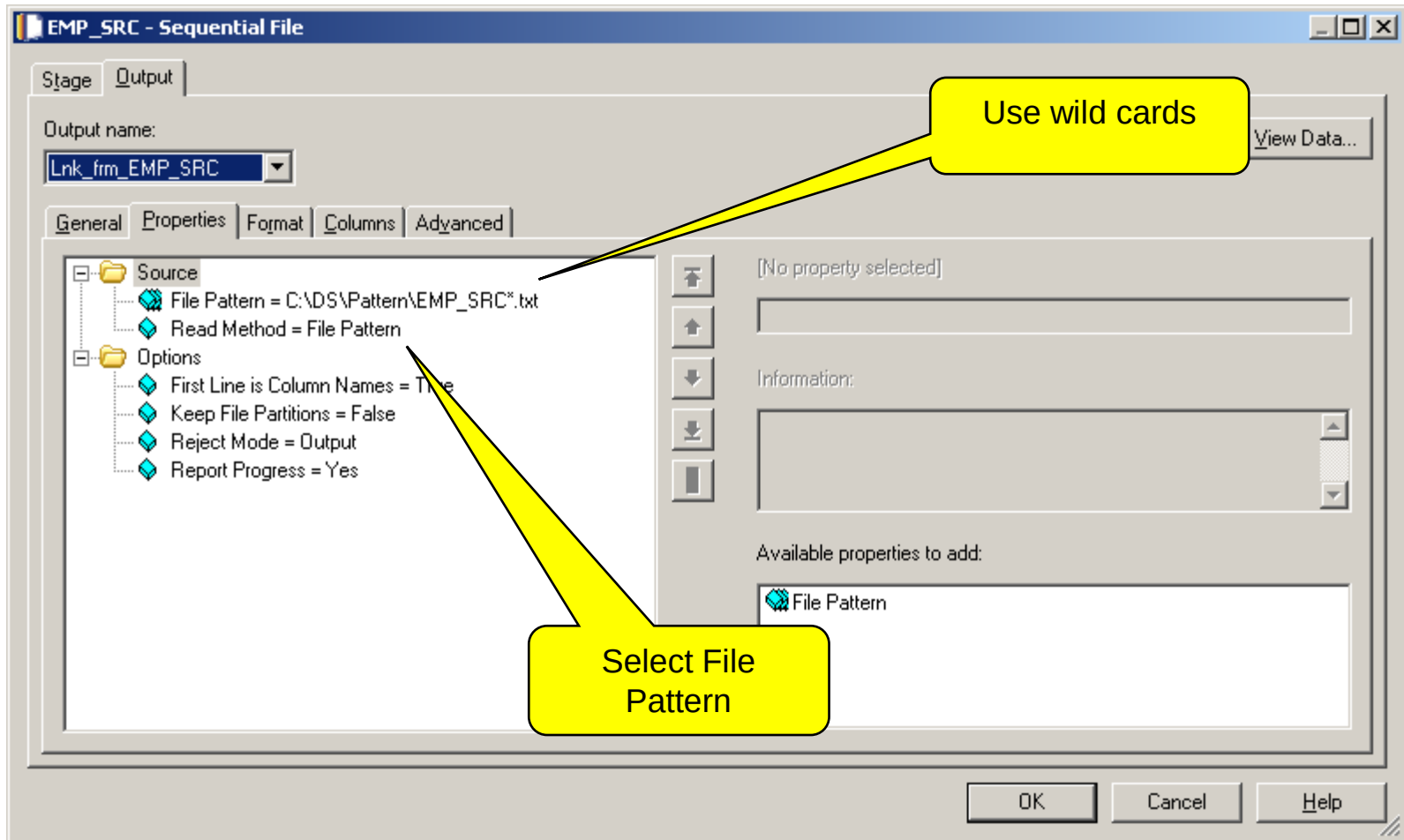
Save as a new Table Definition

☐ Runtime column propagation

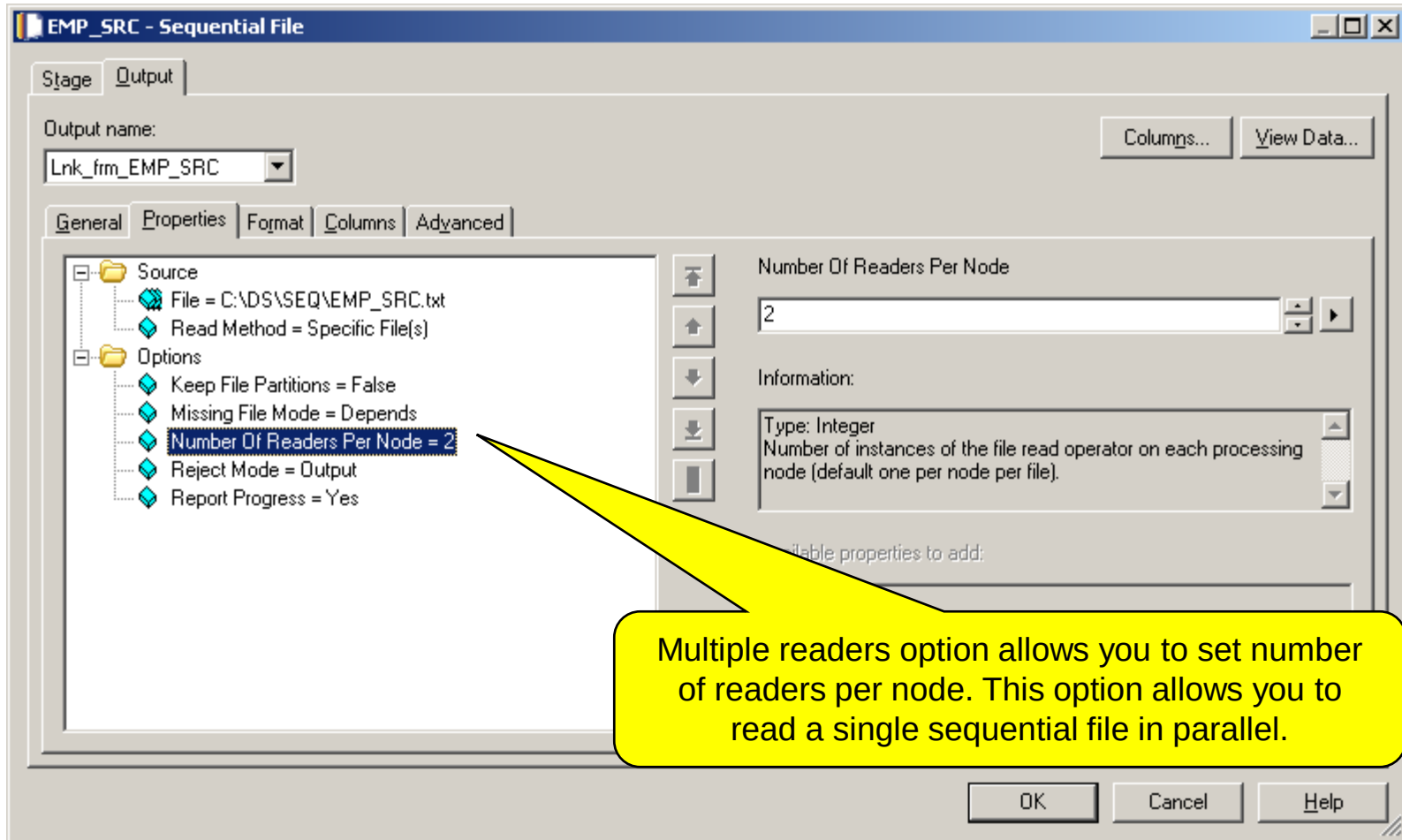
Save... Load...

OK Cancel Help

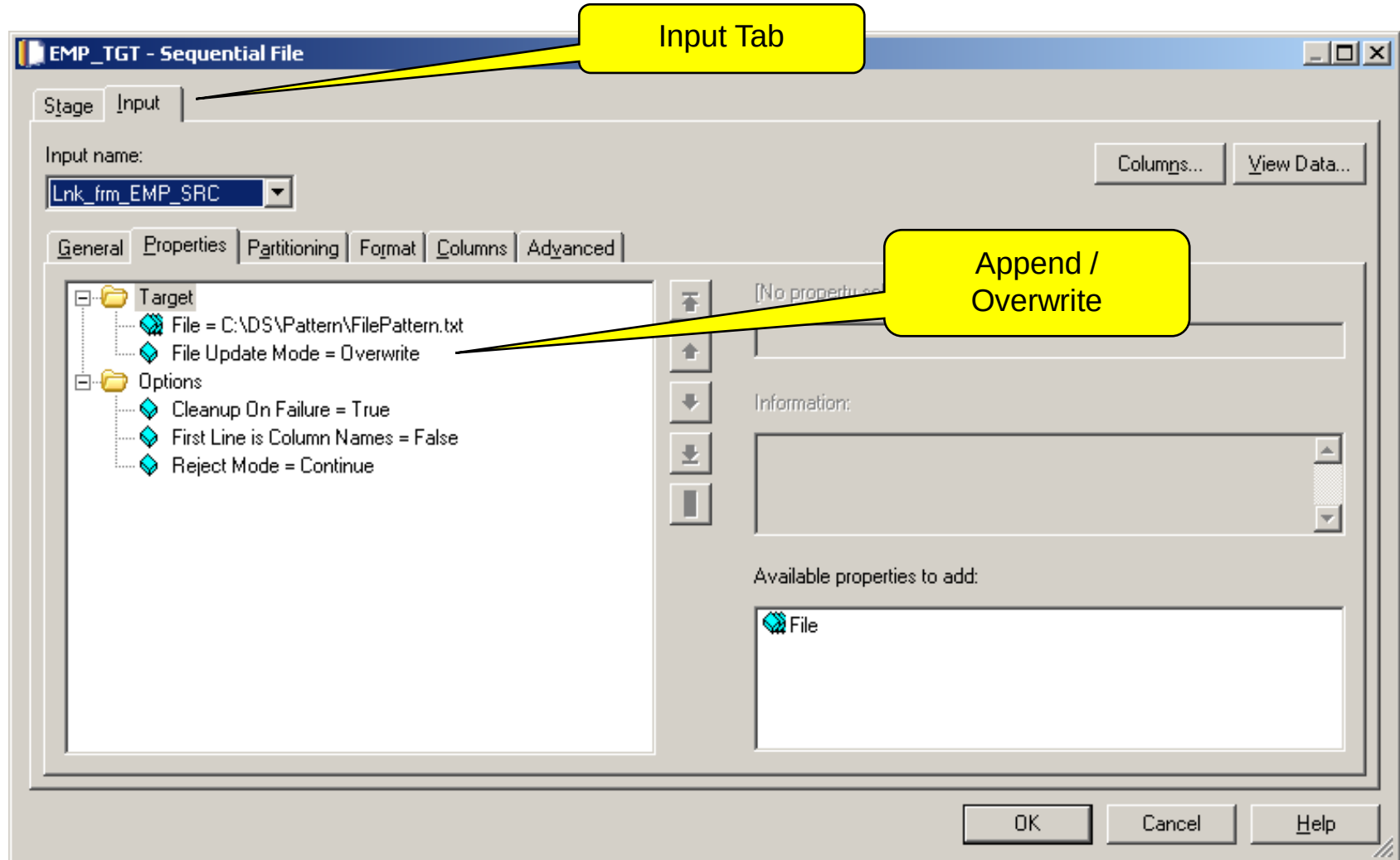
Sequential File Stage - Read Using File Pattern



Sequential File Stage - Multiple Readers Per Node



Sequential File Stage - Target

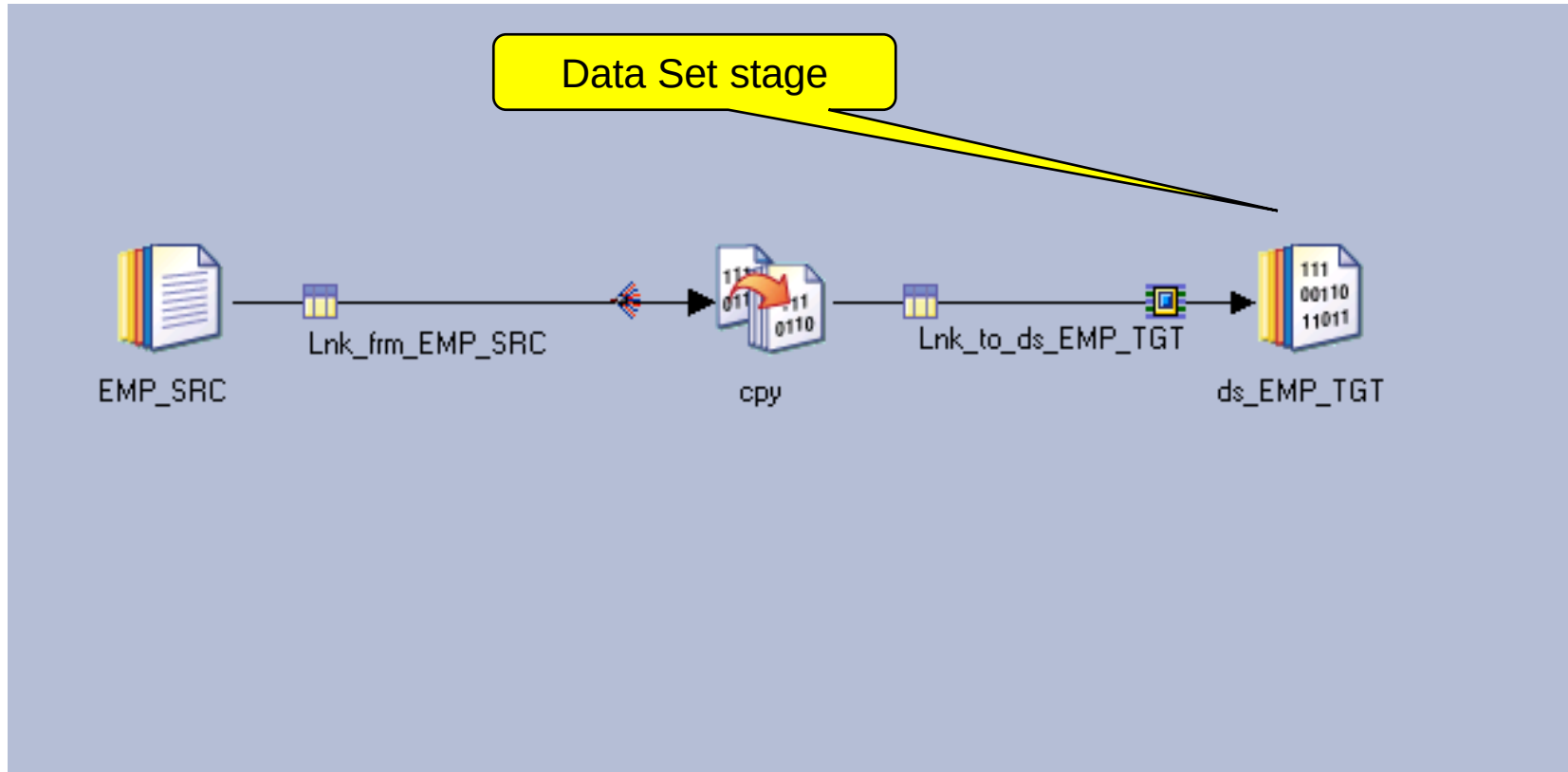


Data Set Stage

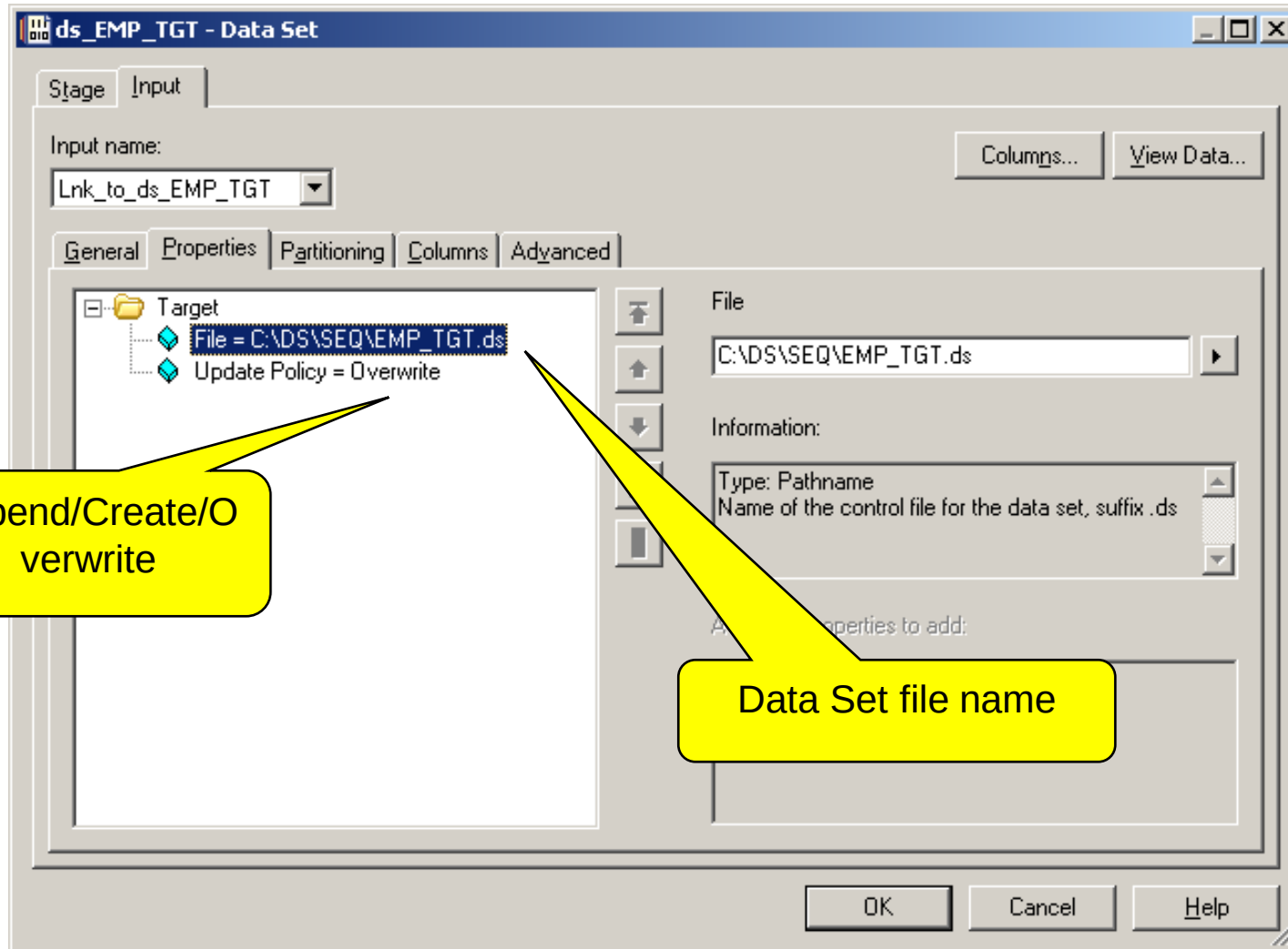


- Binary data file
- Preserves partitioning
 - When reading into a job, data files are read into each partition
- Suffixed by .ds
- Referred to by a header file
- Managed by Data Set Management utility from Designer
- Represents persistent data
- **Key to good performance in set of linked jobs** (job sequence)
 - No import / export conversions are needed
 - No repartitioning needed
- Accessed using Data Set stage
- Implemented with two types of components:
 - Descriptor file:
 - contains metadata, data file locations, but NOT the data itself
 - Data file(s)
 - contain the data
 - multiple files, one per partition (node)

Data Set Stage - Job Example



Data Set Stage - Properties



Data Set Management Utility

Invoke through Tools > Data Set Management from Designer Client

View Data Set
Metadata

The screenshot shows the 'Data Set Management' window with the following sections:

- Data Set Properties:**
 - Name: EMP_TGT.ds
 - Path: C:\DS\SEQ
 - Version: ORCHESTRATE V8.1.0 DM Block Format 6.
 - Created: 2011-03-01 13:01:43
 - Total Records: 14
 - Total 32K Blocks: 2
 - Total Bytes: 7560
- Partitions:**

#	Node	Records	Blocks	Bytes
0	node1	7	1	3780
1	node2	7	1	3780
- Segment Files:**

#	Created	Bytes	Pathname
0	03/01/2011 13:0...	131072	/D=/IBM/InformationServer/Server/Datasets/EMP_TGT.ds.Adr

Buttons at the bottom: Close, Refresh, Output.

Data and Schema View

Data Set Management

Data Set Properties

Name:	EMP_TGT.ds	Total Records:	14
Path:	C:\DS\SEQ	Total 32K Blocks:	2
Version:	ORCHESTRATE V8.1.0 DM Block Format 6.	Total Bytes:	7560
Created:	2011-03-01 13:01:43		

Partitions:

#	Node	Records	Blocks	Bytes
0	node1	7	1	3780
1	node2	7	1	3780

Segment Files:

#	Created	Bytes	Pathname
0	03/01/2011 13:0...	131072	/D=/IBM/InformationServer/Server/Datasets/EMP_TGT.ds.Adm

Close Refresh Output

Click the schema icon to view record schema

Record Schema

```
record
( EMPNO: int32;
  ENAME: string[max=255];
  JOB: string[max=255];
  MGR: int32;
  HIREDATE: date;
  SAL: decimal[10,0];
)
```

☐ Copy to clipboard

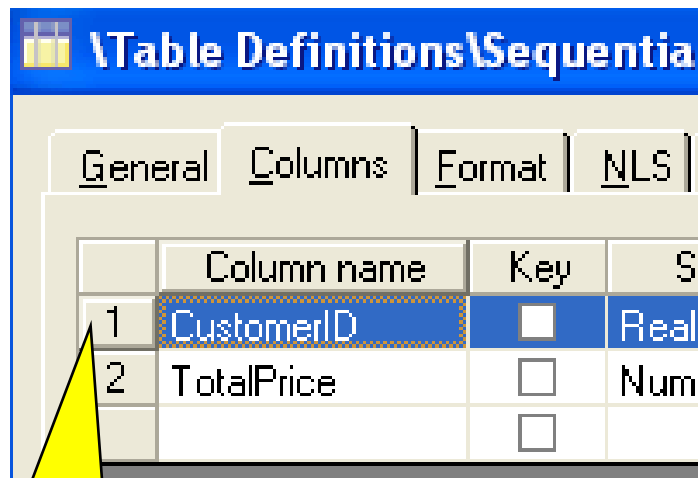
Close

Handling Null Values in Data

- Internally, NULL is represented by a special value outside the range of any existing, legitimate values
- If NULL is written to a non-nullable column, the job will abort
- Columns can be specified as nullable
 - NULLs can be written to nullable columns
- You must “handle” NULLs written to nullable columns in a Sequential File stage
 - You need to tell DataStage what value to write to the file for a NULL
 - Unhandled rows are rejected
 - You specify the value you want to convert to NULL when reading from the file

Specifying a Value for NULL

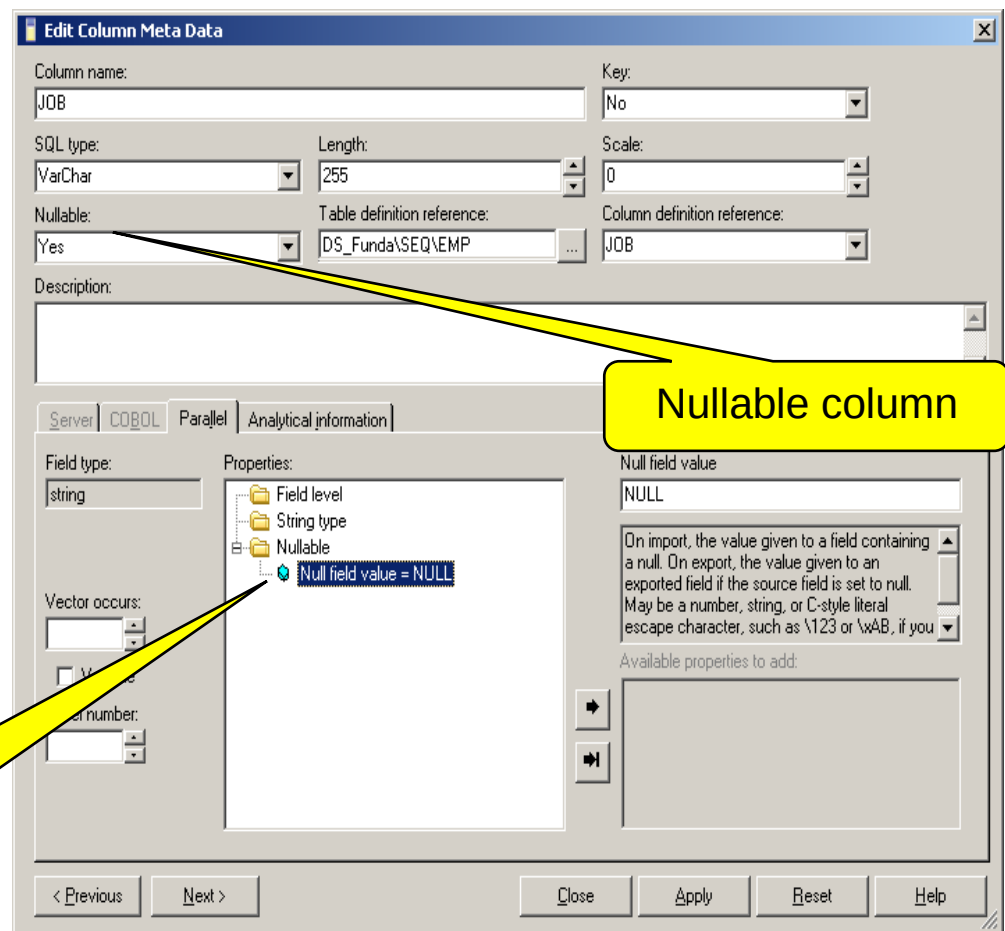
Go to Table Definitions properties -> Columns Tab -> Double click column number



	Column name	Key	S
1	CustomerID	☐	Real
2	TotalPrice	☐	Num

Double click the column number to invoke edit window

Added property



Editable Column Meta Data

Column name: JOB Key: No

SQL type: VarChar Length: 255 Scale: 0

Nullable: Yes Table definition reference: DS_Funda\SEQ\EMP Column definition reference: JOB

Description:

Field type: string

Properties:

- Field level
- String type
- Nullable
 - Null field value = NULL

Vector occurs: 1

Null field value: NULL

On import, the value given to a field containing a null. On export, the value given to an exported field if the source field is set to null. May be a number, string, or C-style literal escape character, such as \123 or \wAB, if you

Available properties to add:

< Previous Next > Close Apply Reset Help

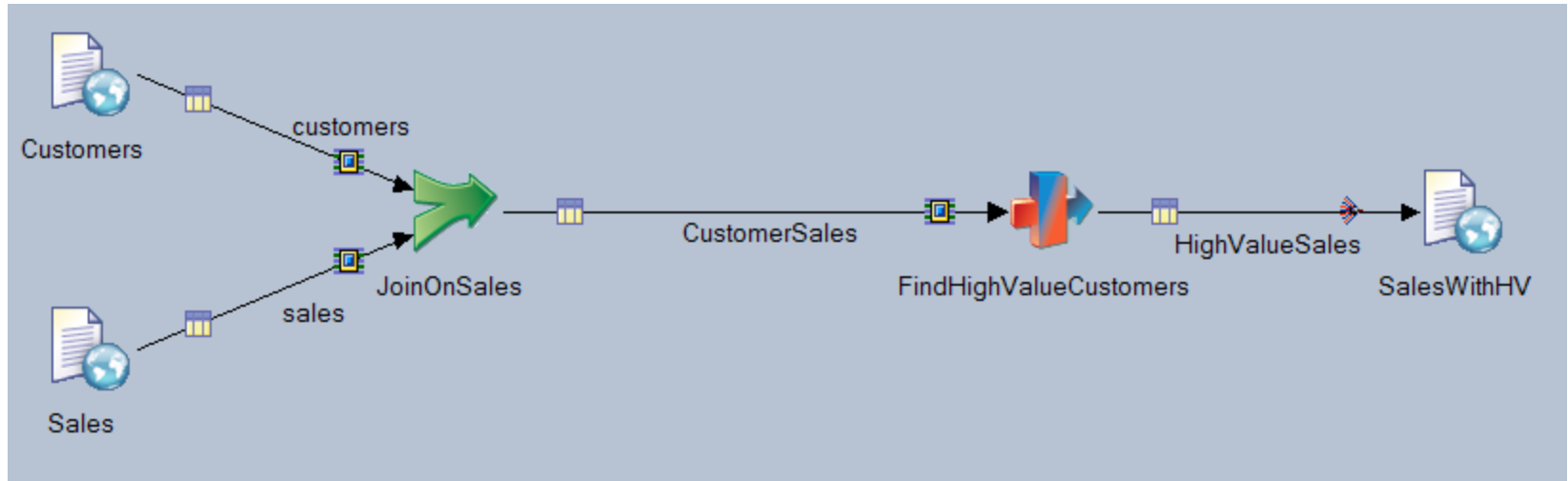
Nullable column

Big Data File Stage (BDFS)



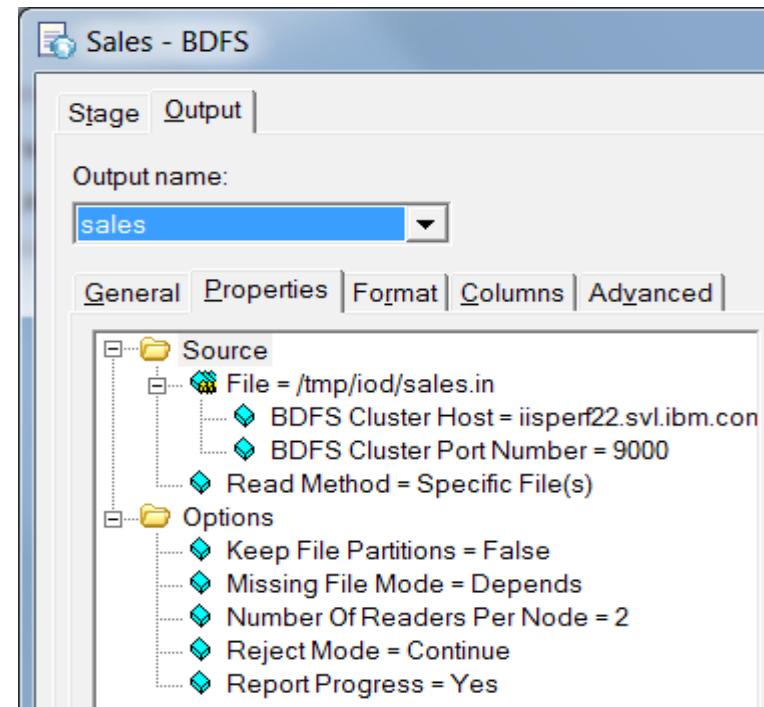
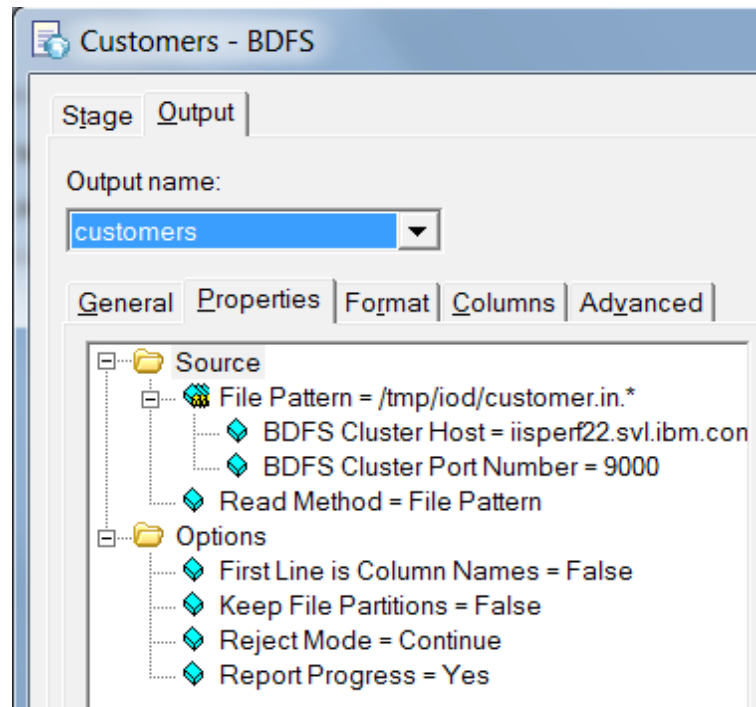
- Provides read/write HDFS file access
- Supported as a patch on top of 8.7
- Embedded to the installation media in 9.1
- Information Server and Hadoop Integration options
- Various and Flexible formatting options (mirrors Sequential File Stage)
- Can read single files in parallel
- Can read multiple files and file patterns in parallel

BDFS Usage Examples



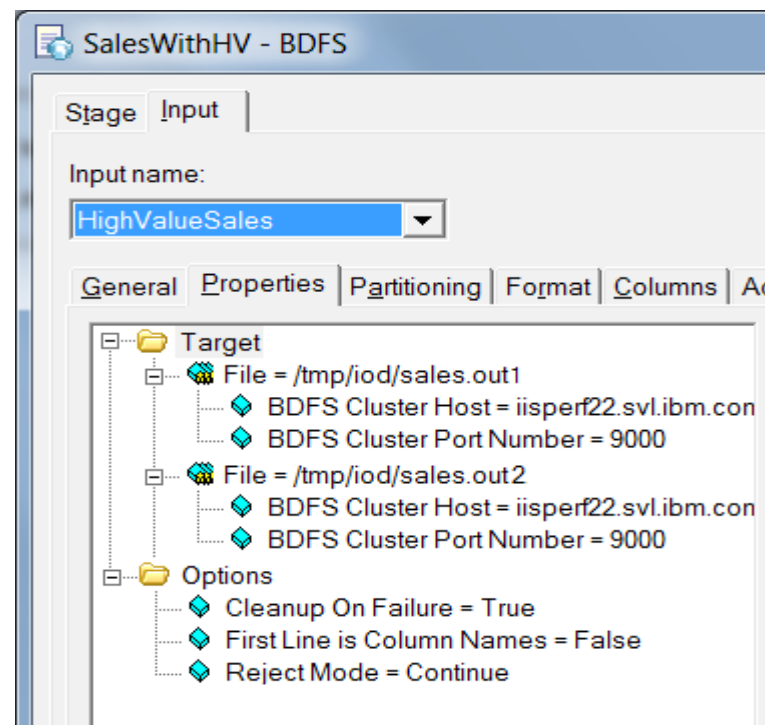
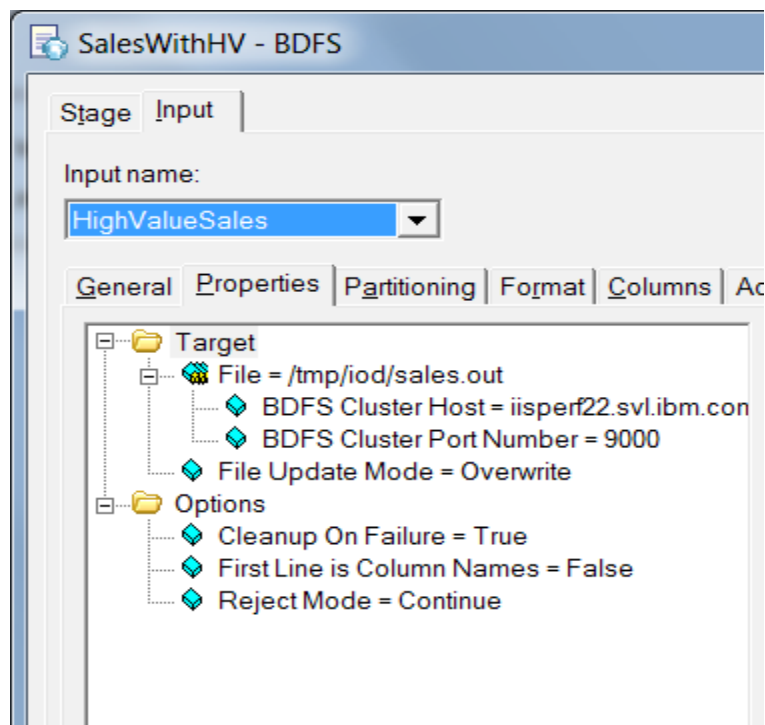
- BDFS can serve as source or target
- Pull data from HDFS to process in DataStage
- Write data to HDFS after processing in DataStage

BDFS As A Source



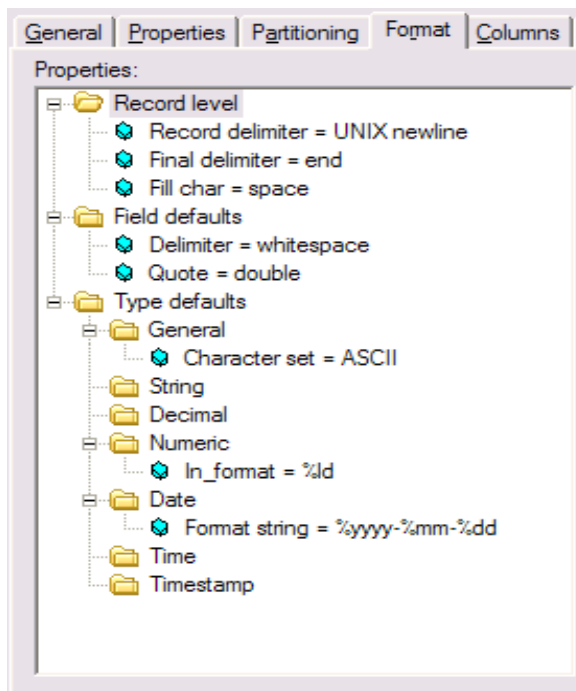
- BDFS introduces new source options
 - [Namenode Cluster Host and Port](#)
- BDFS can read file patterns in parallel
- BDFS can read single files in parallel using Readers option

BDFS As A Target



- BDFS introduces new target options
 - [Namenode Cluster Host and Port](#)
- BDFS usually writes single file
- BDFS supports to write multiple files in parallel

BDFS Formatting Options



- Very flexible consuming and producing binary files and structured text.
- BDFS supports all formatting options in Sequential File Stage
- BDFS supports all types in Sequential File Stage

Expanding on Our Secure and Fast HDFS Connectivity

- File Connector is now supported on all platforms
- BDFS & File Connector support Kerberos secured Hadoop when running natively
- Automatic import / Schema discovery for File Connector
- Data lineage support for File Connector

The image shows two overlapping windows from a data integration tool. The background window is the 'Advanced' tab of a 'Connection' properties dialog, showing configuration for a 'WebHDFS' connection. The foreground window is a 'New Data Connection' dialog, which is currently empty except for the 'Name' field which contains 'Lassen Hadoop HDFS'.

Connection Properties (Advanced Tab):

Property	Value
File system	WebHDFS
Use custom URL	No
Use SSL (HTTPS)	No
Use Kerberos	No
Use keytab	No
Host *	lassen.us.ibm.com
Port	50070
User name *	dsadm
Password *	*****
Keytab *	
Custom URL *	

New Data Connection Dialog:

* Name: Lassen Hadoop HDFS

Description:

* File system: WebHDFS

☐ Use SSL (HTTPS)

☐ Use Kerberos

☐ Use custom URL

* Host: lassen.us.ibm.com

Port:

Buttons: ? OK Cancel

Lab 06: Sequential Data Access (90')